

### What's the key to be efficient in design and manufacturing?

We have an internal design flow called Wipro's EagleVision for silicon and system designs. This helps us to adapt to new technologies, increase automation and enhance productivity. We also have an in-house product compliance lab—Tarang—that helps us provide faster turnaround and get products faster to the market for our customers. We have established strong ecosystem partnerships with IP companies, semiconductor companies and manufacturing companies to increase our efficiency.

### What's important when developing products for different use cases?

Today, semiconductors are bringing 'smartness' into an increasing spectrum of equipment—starting from a child's toy to a complex base station of a wireless network. Across each requirement is a use case, i.e., how the end user would use the product. The form factor, user interface, middleware and silicon all need to be integrated to support and deliver the use case.

This environment is driving a significantly higher collaboration between R&D service providers and semiconductor companies to bring complimentary capabilities to envision the use cases of their joint customers and build platforms and solutions targeted at the use case.

### What interesting market trends have you noticed in India?

Newer technologies are getting adopted faster in India now. We work on many new product developments that are meant to address the specific needs of markets like India. These cut across wireless infrastructure, high-performance computing, healthcare and industrial products.

### In the IT industry, Open Source software has become mainstream and is being increasingly deployed by IT solution providers. Do you see Open Source hardware such as Arduino and Pi Raspberry going mainstream in the electronics design space too?

In the hardware domain, 70 per cent of the engineering cost is in verification. Customers may lose revenue, and market window may be lost due to bugs detected late in the development cycle. Hence we do not expect to see demand for Open Source picking up.

### What technology trends should electronics design engineers keep track of?

Some good segments to work in the electronic design

industry include industrial and automotive microcontrollers, wireless infrastructure segments, consumer digital TV, handheld devices and compute/storage infrastructure. Emerging segments like smart energy product development, medical telemetry application and high-end application processor development too hold a big promise. The top three trends to keep an eye out for are mobility, cloud and analytics. These are sure to revolutionise the electronics design space.

### Do you also make Wipro-branded products?

As part of the PE&MS business, we are into solutions and services for our customers. So we do not have Wipro branded products. We work across the entire spectrum and the eco-system—whether it is semiconductor customers, system companies and service providers—so that we can give an integrated solution to the customer.

As far as semiconductors are concerned, we do the design and development, and chip testing activities locally in India, but the fabrication is handled in other countries through global ecosystem partners. For the systems, we not only do design development in the country but also partner with manufacturing companies for building these

products locally. This way, we can support our customers better during the complete life-cycle of the product. We have a wide network of EMS partners. We work with them to deliver on the manufacturing promise we offer to system companies in India.

### How can one get to work at Wipro?

While hiring a VLSI engineer, we look for strong fundamentals in basic electronics design concepts. We make a lot of investments internally to train and groom these engineers. For lateral hires, we also stress on the need for strong domain understanding or sector-specific knowledge—as the industry is moving towards vertical offerings in chip design.

### What are you doing to bridge the industry-academia gap?

Wipro works closely with the academia and industry-university consortiums. We have played an active role in Aerospace Network Research Consortium (ANRC) with Boeing and IISc, for long-term research in network-centric operations, cognitive radio, IPv6 upgradation and navigation systems. Further, we have university student programmes like India-UK Advanced Technology Centre (IUATC) in IIT Hyderabad, Madras, Bombay and Delhi, RVCE, NIT-K, etc. We also have direct university partnerships with Georgia Tech (USA) and IIIT-B. ●

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